



KUWAIT UNIVERSITY
Faculty of Science
Department of Mathematics

Math 101: Calculus I

First Exam, 22 March 2018

Duration: 90 minutes

- *In questions 1-9, show all your work.*
- *Calculators and communication devices are not allowed in the examination room.*

Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

Q1 (10 pts)	
Q2 (10 pts)	
Q3 (10 pts)	
Q4 (10 pts)	
Q5 (10 pts)	
Q6 (10 pts)	
Q7 (10 pts)	
Q8 (10 pts)	
Q9 (20 pts)	
Total	

1. $[2.5 \times 4 = 10 \text{ pts.}]$ Use the given graph of f to evaluate each of the following limits, if it exists.

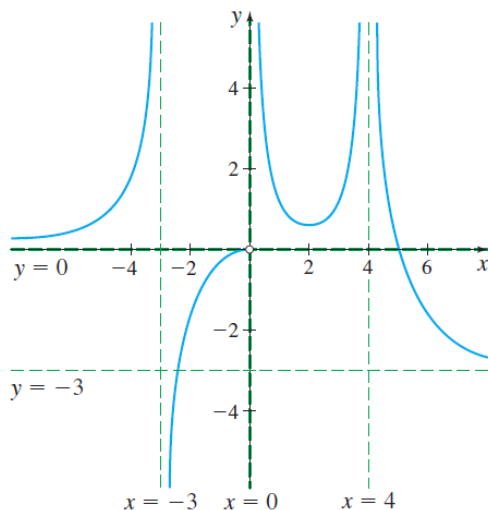


Figure 1: The graph of $y = f(x)$.

(a) $\lim_{x \rightarrow -3} f(x) =$

(b) $\lim_{x \rightarrow 0^-} f(x) =$

(c) $\lim_{x \rightarrow 0^+} f(x) =$

(d) $\lim_{x \rightarrow -\infty} f(x) =$

2. $[5 + 5 = 10 \text{ pts.}]$ The graph of f is given below.

(a) Explain why f is one-to-one.

(b) Find $f^{-1}(2)$.

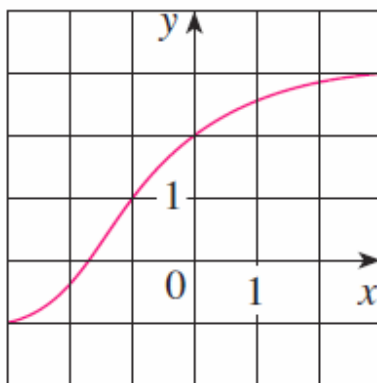


Figure 2: The graph of $y = f(x)$.

3. [10 pts.] Let $f(x) = \ln(2^x - 1)$. Find $f^{-1}(x)$.

4. [2 + 8 = 10 pts.] Define $f(x) = \tan^{-1}\left(\frac{1}{x-2}\right)$ for $x \neq 2$.

(a) Explain why f is discontinuous at $x = 2$.

(b) Classify the type of this discontinuity.

5. [10 pts.] Find a value for the constant A , if any, that makes the function f continuous at $x = 1$, where

$$f(x) = \begin{cases} \frac{\sqrt{x+1} - \sqrt{2x}}{x-1} & \text{if } x \geq 0, x \neq 1 \\ A & \text{if } x = 1. \end{cases}$$

6. [10 pts.] Suppose that the function f is continuous on $[0, 1]$ and satisfies $0 < f(x) < 1$ for all x in $[0, 1]$. Use the Intermediate Value Theorem to show that the equation $f(x) - x = 0$ has a solution in $(0, 1)$.

7. [10 pts.] Let $f(x) = \frac{x^2 - 1}{e^x}$. Find $f'(1)$.

8. [10 pts.] Evaluate the following limit, if it exists.

$$\lim_{x \rightarrow 0} \left(\frac{\frac{1}{x^2 + 4} - \frac{1}{4}}{x} \right).$$

9. [10+10 = 20 pts.] Let $f(x) = \frac{x|x+1|}{x^2 - 1}$.

(a) Find the horizontal asymptotes of the graph of f , if any.

(b) Find the vertical asymptotes of the graph of f , if any.

Solutions

1. $[2.5 \times 4 = 10 \text{ pts.}]$

(a) $\lim_{x \rightarrow -3} f(x) = \boxed{\text{DNE}}$. (It is clear that $\lim_{x \rightarrow -3^+} f(x) = -\infty$ and $\lim_{x \rightarrow -3^-} f(x) = \infty$).

(b) $\lim_{x \rightarrow 0^-} f(x) = \boxed{0}$.

(c) $\lim_{x \rightarrow 0^+} f(x) = \boxed{\infty}$.

(d) $\lim_{x \rightarrow -\infty} f(x) = \boxed{0}$.

2. $[5 + 5 = 10 \text{ pts.}]$

(a) Any horizontal line intersects the graph of f at most once. Therefore, the function is one-to-one by the horizontal line test.

(b) $\boxed{f^{-1}(2) = 0 \text{ since } f(0) = 2}$.

3. $[10 \text{ pts.}]$ Let $f(x) = \ln(2^x - 1)$.

1. Write: $\boxed{y = \ln(2^x - 1)}$.

2. Solving the above equation for x in terms of y , we get: $\boxed{x = \log_2(e^y + 1)}$.

3. Interchanging x and y , we get: $\boxed{y = \log_2(e^x + 1)}$.

Therefore, $\boxed{f^{-1}(x) = \log_2(e^x + 1)}$.

4. $[2 + 8 = 10 \text{ pts.}]$ Define $f(x) = \tan^{-1}\left(\frac{1}{x-2}\right)$ for $x \neq 2$.

(a) It is clear that $\boxed{f(2) \text{ is undefined}}$. Thus, f is discontinuous at $x = 2$.

(b) Since $\boxed{\lim_{x \rightarrow 2^\pm} f(x) = \pm \frac{\pi}{2}}$, the function $\boxed{f \text{ has a jump discontinuity at } x = 2}$.

5. $[10 \text{ pts.}]$

$$f(x) = \begin{cases} \frac{\sqrt{x+1} - \sqrt{2x}}{x-1} & \text{if } x \geq 0, x \neq 1 \\ A & \text{if } x = 1. \end{cases}$$

It is clear that $\boxed{f(1) = A}$ and

$$\begin{aligned} \lim_{x \rightarrow 1} f(x) &= \lim_{x \rightarrow 1} \frac{\sqrt{x+1} - \sqrt{2x}}{x-1} \\ &= \lim_{x \rightarrow 1} \frac{\sqrt{x+1} - \sqrt{2x}}{x-1} \times \frac{\sqrt{x+1} + \sqrt{2x}}{\sqrt{x+1} + \sqrt{2x}} \\ &= \lim_{x \rightarrow 1} \frac{1-x}{(x-1)(\sqrt{x+1} + \sqrt{2x})} \\ &= \lim_{x \rightarrow 1} \frac{-1}{\sqrt{x+1} + \sqrt{2x}} = \frac{-1}{2\sqrt{2}} \end{aligned}$$

The function is continuous at $x = 1$ if $f(1) = \lim_{x \rightarrow 1} f(x) = \frac{-1}{2\sqrt{2}}$, i.e., $A = \frac{-1}{2\sqrt{2}}$.

6. [10 pts.] Define $g(x) = f(x) - x$. It is clear that the function g is continuous on $[0, 1]$ since it is the difference of two continuous functions on $[0, 1]$. Also since $0 < f(x) < 1$, we have $g(0) = f(0) - 0 = f(0) > 0$ and $g(1) = f(1) - 1 < 0$. Therefore, by the IVT there exists a c in $(0, 1)$ such that $g(c) = f(c) - c = 0$ or equivalently $f(c) = c$. Thus c is a solution for $f(x) - x = 0$.

7. [10 pts.] Let $f(x) = \frac{x^2 - 1}{e^x}$.

We have $f'(x) = \frac{(2x)e^x - e^x(x^2 - 1)}{e^{2x}}$. Therefore, $f'(1) = \frac{2e}{e^2} = \frac{2}{e}$.

8. [10 pts.] $\lim_{x \rightarrow 0} \left(\frac{\frac{1}{x^2 + 4} - \frac{1}{4}}{x} \right) = \lim_{x \rightarrow 0} \left(\frac{\frac{-x^2}{4(x^2 + 4)}}{x} \right) = \lim_{x \rightarrow 0} \frac{-x}{4(x^2 + 4)} = 0$.

9. [10+10 = 20 pts.] Let $f(x) = \frac{x|x+1|}{x^2 - 1}$.

(a) We have $\lim_{x \rightarrow \pm\infty} \frac{x|x+1|}{x^2 - 1} = \lim_{x \rightarrow \pm\infty} \frac{\pm x(x+1)}{x^2 - 1} = \lim_{x \rightarrow \pm\infty} \frac{\pm(x^2 + x)}{x^2 - 1} = \pm 1$.

Therefore, the lines $y = \pm 1$ are horizontal asymptotes to the curve $y = f(x)$.

(b) We have $\lim_{x \rightarrow 1^\pm} \frac{x|x+1|}{x^2 - 1} = \lim_{x \rightarrow 1^\pm} \frac{x(x+1)}{(x-1)(x+1)} = \pm\infty$.

Note that $\lim_{x \rightarrow -1^\pm} \frac{x|x+1|}{x^2 - 1} = \lim_{x \rightarrow -1^\pm} \frac{\pm x(x+1)}{(x-1)(x+1)} = \lim_{x \rightarrow -1^\pm} \frac{\pm x}{(x-1)} = \pm 1/2$.

Therefore, the line $x = 1$ is a vertical asymptote to the curve $y = f(x)$.